

CV

NAVAKANTH RAO VAIKAR

Translational Oncology Research Scientist

Precision Medicine | Patient-Derived Organoids (PDOs) | Preclinical Oncology

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PROFESSIONAL SUMMARY

Translational oncology researcher with 12+ years of experience in cancer pharmacology, precision medicine, and preclinical drug development. Specialized in Patient-Derived Organoid (PDO) platforms, Triple Negative Breast Cancer (TNBC), colorectal cancer models, molecular pharmacology, and translational drug screening strategies. Experienced in bridging bench-to-bedside oncology research through clinically relevant in vitro and in vivo models, biomarker-driven therapeutic evaluation, and multidisciplinary collaborations. Proven record of international presentations at AACR and EORTC with multiple peer-reviewed publications in oncology and pharmacology.

CORE EXPERTISE

- Patient-Derived Organoid (PDO) Culture & Drug Screening
- Translational Oncology Research
- Precision Medicine Strategies
- Triple Negative Breast Cancer (TNBC) Models
- Colorectal Cancer (CRC) Xenograft Models
- Preclinical Drug Evaluation
- Antibody-Drug Conjugates (ADCs)
- Molecular Pharmacology
- Flow Cytometry & Immunohistochemistry
- Biomarker-Driven Therapeutics

PROFESSIONAL EXPERIENCE

Senior Research Assistant

National Taiwan University, Taiwan | 2024–2025

- Investigated anti-cancer therapeutic activity of ecteinascidins in SARS-CoV-2 infected TNBC models.
- Conducted translational oncology studies integrating molecular pharmacology and precision medicine approaches.
- Supported preclinical therapeutic evaluation using clinically relevant cancer models.

Ph.D. Candidate & Research Assistant

Academia Sinica & Tzu Chi University, Taiwan | 2017–2023

- Developed and optimized Patient-Derived Organoid (PDO) platforms for colorectal cancer (CRC) and Triple Negative Breast Cancer (TNBC) translational research.
- Conducted preclinical drug screening using clinically relevant PDO and xenograft models to evaluate therapeutic efficacy across CRC consensus molecular subtypes (CMS).
- Investigated dual-functional anti-cancer compounds targeting angiogenesis and DNA synthesis pathways.
- Performed in vitro and in vivo pharmacological studies including flow cytometry, histopathology, molecular characterization, and biomarker analysis.
- Collaborated within multidisciplinary oncology research teams focused on precision medicine and translational therapeutic development.

Research Assistant

Hualien Armed Forces General Hospital & National Dong Hwa University, Taiwan | 2015–2017

- Supported translational pharmacology research focused on molecular mechanisms and therapeutic evaluation in disease models.

Senior Research Associate

Quest Life Sciences Pvt. Ltd., India | 2013–2015

- Contributed to preclinical research and drug discovery activities within multidisciplinary research programs.

SELECTED ACHIEVEMENTS

- Presented oncology research at AACR Annual Meetings (USA) and EORTC Molecular Targets Symposium (Spain).
- Developed PDO-based translational screening platforms for CRC and TNBC therapeutic evaluation.
- Published multiple peer-reviewed studies in oncology and pharmacology journals.
- Experience in clinically relevant preclinical cancer models and precision medicine strategies.

SELECTED ONCOLOGY PUBLICATIONS

- Rao VN, et al. A novel hybrid with anti-angiogenic and DNA crosslinking activities potently suppresses colorectal cancer cells and patient-derived organoids in mice. *Cancer* (Under Review), 2026.
- Rao VN, et al. A dual functional activity subdues the growth of consensus molecular subtypes 1 and 4 in colorectal cancer xenografts. *Cancer Research*, 2022.
- Rao VN, et al. A dual functional agent inhibits angiogenesis and DNA synthesis resulting in growth suppression of consensus molecular subtypes 1 and 4 in colorectal cancer xenografts. *Cancer Research*, 2021.
- Rao VN, et al. Design, synthesis and antitumour evaluation of pyrrolo derivatives. *European Journal of Medicinal Chemistry*, 2020.

INTERNATIONAL CONFERENCES

- AACR Annual Meeting – San Diego, USA (2021)
- AACR Annual Meeting – New Orleans, USA (2022)
- EORTC-NIC-AACR Symposium – Barcelona, Spain (2023)
- Asia Pacific Federation of Pharmacologists (APFP), 2023

EDUCATION

Ph.D. Program in Pharmacology & Toxicology (Requirements Fulfilled)

Academia Sinica & Tzu Chi University, Taiwan | 2017–2023

M.Sc. Biotechnology

Periyar University & RGCB Trivandrum, India | 2010–2012

HONORS & AWARDS

- TIGP Travel Award – Best Poster, Academia Sinica (2020)
- Best Researcher Award – IIT Guwahati, India (2017)
- Dream Player Award – Academia Sinica (2022)

References

Dr. Te-Chang Lee

Adjunct Research Fellow, Institute of Biomedical Sciences, Academia Sinica

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Dr. Jean Marc Egly

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